# KARTIK SONI

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## EDUCATION

**B.S. Computer Science (Honors) · University of Massachusetts Amherst (UMass Amherst) · GPA 3.9/4.0 Aug 2019 - May 2023**

## AWARDS & RECOGNITION

Cum Laude (Top 20%) · Dean's List (Multiple Semesters) · Bay State Scholarship · Employee of the Month & Team Awards @ Whova

## TECHNICAL SKILLS

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|---|---|
| **AI / ML** | PyTorch · TensorFlow/Keras · Transformers· HuggingFace · LLMs · Fine-tuning · CNNs · NLP · Computer Vision OpenCV · Naïve Bayes · Cross-validation · Hyperparameter Tuning |
| **Backend** | Spring Boot · Java · Python · Django · Node.js · REST APIs · Redis · Microservices · JUnit · Mockito · Swagger |
| **Frontend** | Angular · React.js · TypeScript · JavaScript · HTML/CSS · MVVM · Lazy Loading · Reactive Forms |
| **Practices** | System Design · HLD/LLD · Agile/Scrum · Design Patterns · Code Review · PostgresSql |

## EXPERIENCE

**Software Engineer** · Whova *San Diego, CA, USA | Aug 2023 - May 2026*

- Owned POC coding, MVP scoping, Epic estimation, stakeholder communication, and cross-team delivery for an event management platform serving large-scale bookings.
- Worked on attendee engagement features such as Name Badges, Certificates, Post Event Report, Call For Speakers, and feature based referral program.
- Developed core features for MicroEvents, a six month, company-wide initiative, collaborating across engineering teams, designers, and product managers to launch a subscription-based event management system.
- Built an AI agent with a tool-calling feedback loop (DB queries, web search, email) on a self-hosted Llama/vLLM/KServe stack deployed on GKE, with capabilities exposed as reusable MCP servers and served via FastAPI.
- Re-architected a high-traffic Angular web app with MVVM, lazy loading to reduce initial load time (from 5 to 2 seconds), and OnPush change detection to improve runtime performance significantly.
- Designed robust REST APIs using Django and ensured high system reliability to deliver seamless, high-performance web applications.
- Recognised as Employee of the Month for engineering innovation; mentored 3 interns through onboarding, Agile practices, and project bootstrapping.
- Optimised legacy code for multiple features, reducing DB load and celery backlog queue.

**ML Researcher** · UMass Amherst - Prof. Jaime Davila *Amherst, MA, USA | Feb 2023 - May 2023*

- Designed and trained CNN architectures for medical image classification (disease detection) on datasets of 4 000 - 13 000 images, iterating on kernel sizes, depth, and regularisation to combat overfitting.
- Applied 4-fold cross-validation and systematic hyperparameter search (learning rate, depth, optimiser), switching to SGD and achieving a final F1 score of 91%.
- Improved dataset quality by sourcing diverse, well-distributed images and applying augmentation (rotations, flips, brightness/contrast shifts), boosting generalisation on held-out sets.

**Teaching Assistant - Systems Programming** · UMass Amherst *Amherst, MA, USA | Sep 2022 - May 2023*

- Built a transformer-based grading system using BERT + sentence-transformers to evaluate semantic similarity, extract key concepts (NER), and generate feedback for a Systems Programming course. Achieved 92% agreement with human graders across 280+ assignments, reducing grading time from 4 hours/week to 20 minutes/week.
- Supported 100+ students on concurrency, synchronisation, pointers, dynamic memory, and control flow; designed assignments and labs that consistently received positive feedback.

**Software Engineer Intern** · ISO New England *Holyoke, MA, USA | Jun 2022 - Aug 2022*

- Migrated a legacy C++ application to a Spring Boot Java microservice; added JWT-based auth, role-based access, and Redis caching, reducing average API latency noticeably.
- Integrated SLF4J + Sentry for structured error logging; applied Lombok and SonarLint to raise code quality; scheduled automated emails via SendGrid cron jobs.

## HOBBY PROJECTS

**Neural Network Training Observability Tool** · NumPy · PyTorch

- Built a per-layer training observability tool in PyTorch using forward hooks that detects exploding/vanishing gradients/activations and dead neurons from layer statistics (activation mean/std, dead fraction, gradient norms) and recommends targeted fixes - LeakyReLU, weight re-initialization, learning rate reduction - based on each failure's layer-wise signature.
- Closed the loop on MNIST digit classification and sentiment analysis (learned embeddings) by deliberately inducing each pathology, confirming correct diagnosis, applying the recommended fix, and verifying training recovery.
- Developed a from-scratch NumPy reference implementation of the complete training stack - forward pass + backpropagation via chain rule, gradient descent, linear regression through multi-layer perceptron, MSE and categorical cross-entropy losses, sigmoid/ReLU/softmax activations - verified with finite-difference gradient checks.
- Implemented the modern training stabilizer toolkit: Xavier/Kaiming weight initialization, Dropout, and the full normalization family (BatchNorm with running statistics and train/eval modes, LayerNorm, RMSNorm), demonstrating the statistical invariants each enforces.

**LLM Domain Adaptation & RAG Pipeline ·** Python · HuggingFace Transformers · LangChain · FAISS

- Fine-tuned an open-source transformer (LLaMA / Mistral) using LoRA/QLoRA for domain-specific Q&A; reduced hallucination rate and improved answer relevance on domain benchmarks.
- Built a retrieval-augmented generation (RAG) pipeline with LangChain and FAISS vector store for document-grounded responses, enabling accurate answers over private knowledge bases.
- Evaluated outputs with BLEU, ROUGE, and human-preference scores; optimised prompt templates and chunk-size strategies to balance latency and accuracy.

**Smart Home IoT System** · Raspberry Pi · Python · Node.js · Angular · PostgreSQL

- Built a Raspberry Pi temperature/humidity monitor (DHT11) with sensor data streamed to Supabase PostgreSQL via a Node.js API; deployed an Angular dashboard on Vercel for real-time room visualisation and remote relay control.
- Added Pi-hole DNS-level ad blocking and relay-based scheduled lighting control, demonstrating edge AI/IoT system design end-to-end.

**Real-Time Mask Detection (CNN)** · TensorFlow · Keras · OpenCV · Python

- Trained a CNN on a manually labelled, augmented dataset (rotations, flips, brightness/contrast shifts, zoom) to achieve 92% real-time mask-detection accuracy.
- Deployed as an open-source initiative during COVID using OpenCV for live video inference; contributed to public safety tooling adopted by multiple organisations.

## RELEVANT COURSE WORK

AI – Machine Learning, Computer Vision, Data Structures & Algorithms, NLP, Search Engine, Operating Systems, Computer Network, Programming Methodology, Reasoning under uncertainty